

NITHIN POLAVARAPU

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, AI & ML) with hands-on experience building end-to-end ML pipelines, multi-modal AI systems, and data-driven applications in Python. Completed two ML internships and shipped two full-stack AI projects spanning predictive modeling, NLP, computer vision, and healthcare diagnostics. Targeting entry-level roles in AI Engineering, Machine Learning, or Python development.

TECHNICAL SKILLS

Languages:	Python, C
ML / DL:	Scikit-Learn, TensorFlow, Keras, PyTorch, Random Forest, Logistic Regression, CNN, Supervised Learning
AI / NLP:	DistilBERT, Hugging Face Transformers, OpenAI CLIP, Multi-Modal AI, Predictive Modeling, Prompt Engineering
Data:	NumPy, Pandas, Streamlit, Matplotlib
Databases:	MongoDB, MySQL, PostgreSQL
Dev & Tools:	Git, GitHub, Vercel, REST APIs, Authentication, Caching, Error Resilience, SDLC

EXPERIENCE

Machine Learning Intern — CodTech IT Solutions - Designed scalable ML pipelines to ingest and process structured and semi-structured enterprise datasets. - Built and deployed predictive models that reduced manual data-analysis cycles for business stakeholders. - Developed an AI-driven analytics suite integrating ML models directly into business intelligence workflows.	Jan 2026 – Apr 2026
Machine Learning Intern — Micro Information Technology Services - Applied supervised learning algorithms to real-world classification tasks, improving prediction accuracy. - Engineered features from raw datasets that measurably lifted model performance on held-out test sets. - Conducted exploratory data analysis on operational datasets to surface patterns for business decisions.	May 2025 – Jun 2025

PROJECTS

Intelligent Business Analytics & Automation Suite - Built a unified Streamlit dashboard combining predictive analytics, NLP, and computer vision in a single deployable application. - Trained a Random Forest dash model on a 7,043-record enterprise dataset, achieving 80% churn-prediction accuracy and automating customer risk assessment. - Integrated DistilBERT across 15,000 features for semantic sentiment analysis, reaching 99% accuracy at under 3-second processing latency. - Deployed an OpenAI CLIP zero-shot vision system on 70,000 images, achieving 98% tag-alignment accuracy for automated inventory classification.	Streamlit • DistilBERT • CLIP • Random Forest
MediAI — Multi-Disease Screening Platform (4 Diseases) - Built a full-stack healthcare screening platform covering four diseases across tabular clinical data and medical imaging pipelines. - Developed Random Forest & Logistic Regression models on 760+ patient records (Diabetes) and 400+ clinical samples (Chronic Kidney Disease), achieving 85%+ diagnostic accuracy. - Trained CNNs on 5,800+ chest X-rays (Lung Disease) and 7,800+ ultrasound/histopathology images (Breast Cancer) for automated binary classification. - Architected modular Streamlit UI with dual-database support (MongoDB + SQL), secure authentication, PDF report generation, and failure-recovery systems.	TensorFlow • Keras • Scikit-Learn • MongoDB • SQL
Neural Language Model — Next-Word Prediction Built a next-word prediction model from scratch: tokenization, word embeddings, iterative training, and an end-to-end inference pipeline.	Python • PyTorch

EDUCATION

B.Tech, Computer Science Engineering (AI & ML) — Parul University, Gujarat CGPA: 7.76	Jun 2022 – May 2026
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CERTIFICATIONS

Supervised Machine Learning: Regression & Classification	DeepLearning.AI / Stanford — Coursera, 2025
Google Prompting Essentials	Google — Coursera, 2025